# Folding the Neumann Laplacian with Argyris elements: the square works, the pizza slice does not

August 19, 2026

This note is the computational companion to `colbrook_setup.tex`. It carries out case 3 of that note (Laplace folding, $V = H^2 \cap H^1$, $H = L^2$) for the *Neumann* Laplacian, so that $V = H^2(\Omega)$ and no Dirichlet trace has to be pinned down on the Argyris degrees of freedom. Code: `Codes/folded_neumann_argyris.py`, `Codes/neumann_exact_spectra.py`.

## 1 The discrete problem

Let $A_N$ be the Neumann Laplacian,

$$A_N = -\Delta, \qquad \mathrm{D}(A_N) = \{u \in H^1(\Omega) : \Delta u \in L^2(\Omega),\ \partial_\nu u = 0\} \subset L^2(\Omega), \tag{1}$$

which is self-adjoint, so $\gamma(z, A_N) = \sigma_{\inf}(A_N - z)$ and folding computes $\mathrm{dist}(z, \sigma(A_N))$ for real $z$.

We do *not* discretise $A_N = M^{-1}L$ directly. Following the variational viewpoint of `colbrook_setup.tex` we keep the weak Laplacian $L\colon H^1 \to (H^1)^*$, $\langle Lu, w\rangle = (\nabla u, \nabla w)$, and the mass map $M\colon L^2 \to (L^2)^*$, and we fold

$$G(z) = (L - zM)^* M^{-1} (L - zM) : V \to V^*, \qquad \sigma_{\inf}(A_N - z)^2 = \inf_{u \in V \setminus \{0\}} \frac{\|(L - zM)u\|^2_{(L^2)^*}}{\|u\|^2_{L^2}}, \tag{2}$$

where $\|r\|_{(L^2)^*} = \sup_{w \in H^1} \langle r, w\rangle / \|w\|_{L^2}$.

**The boundary condition is natural, and $M^{-1}$ is what enforces it.** Because $V_h$ is $C^1$-conforming we really do fold $\Delta = \mathrm{div}\,\mathrm{grad}$: for $u \in H^2(\Omega)$ the integration by parts is exact,

$$\langle (L - zM)u, w\rangle = (\nabla u, \nabla w) - z(u, w) = (-\Delta u - zu,\, w)_{L^2(\Omega)} + (\partial_\nu u,\, w)_{L^2(\partial\Omega)}, \tag{3}$$

and the code uses the second spelling, so that $\mathrm{div}(\mathrm{grad}(u))$ and the boundary term both appear explicitly. That surface term carries no mesh size and no parameter — it is not a penalty and not Nitsche. It *is* the Neumann condition, as the following three-way comparison on the unit square ($N = 8$, exact values 3.0000000, 2.1303956, 9.4784176) shows:

| form used for $\langle (L - zM)u, \psi\rangle$ | $z = 3$ | $z = 12$ | $z = 30$ |
|---|---|---|---|
| $(\nabla u, \nabla\psi) - z(u, \psi)$ | 3.0000000 | 2.1303956 | 9.4784179 |
| $-(\Delta u, \psi) + (\partial_\nu u, \psi)_{\partial\Omega} - z(u, \psi)$ | 3.0000000 | 2.1303956 | 9.4784179 |
| $-(\Delta u, \psi) - z(u, \psi)$ (surface term dropped) | $10^{-7}$ | no convergence | |

The first two rows agree to every digit; the third is the maximal Laplacian. Note that the integration by parts may only be performed on the Argyris slot: $w \in W_h$ lives in $H^1$ only, so $\mathrm{div}(\mathrm{grad}(w))$ is illegal. Returning to (3), so $\|(L - zM)u\|_{(L^2)^*} = \|\Delta u + zu\|_{L^2}$ when $\partial_\nu u = 0$ and $+\infty$ otherwise. In other words the requirement "$(L - zM)u$ must be representable in $L^2$", which is exactly the requirement that $M^{-1}$ can be applied, *is* the Neumann condition. Nothing has to be imposed on the Argyris dofs, strongly or weakly.

This matters. If instead one folds the strong form and minimises $\int_\Omega |\Delta u + zu|^2$ over all of $\mathrm{Argyris}_h$ with no constraint, one is folding the *maximal* Laplacian, whose spectrum is all of $\mathbb{C}$; numerically $\sigma_{\inf} \approx 0$ for every $z$ (on the unit square with $N = 16$ we measured $\sigma_{\inf} \le 3.6 \cdot 10^{-5}$ at $z = 1, 5, 12, 30$, decreasing under refinement). And if one imposes $\partial_\nu u_h = 0$ pointwise on the Argyris dofs one gets the correct answer on a square, but on a domain whose boundary polygon has corner angles tending to $\pi$ the constraint forces $\nabla u_h = 0$ at every boundary vertex and the space collapses: on a regular $n$-gon inscribed in the unit disc we measured, at $z = 3$ and $z = 7$, the values $(1.93, 2.07)$ for $n = 4$, $(2.56, 4.05)$ for $n = 8$, $(3.00, 6.06)$ for $n = 16$ and $(3.0000, 7.0000)$ for $n = 32$ — i.e. $\sigma_{\inf}(z) \to |z|$, only constants survive. This is the Babuška paradox in its usual disguise. The dual-norm form above has neither problem.

**Discretisation, and where the $M^{-1}$ goes.** $V_h = \mathrm{Argyris}_5(\mathcal{T}_h) \subset H^2$, and the supremum in the dual norm is taken over $W_h = \mathrm{CG}_4 \subset H^1$. The test space must be genuinely richer than the trial space, $\dim W_h > \dim V_h$: otherwise the residual matrix $R(z) = K - zM_{WV}$ has a kernel and $\sigma_{\inf} \equiv 0$.

Unlike `folded_maxwell.py`, where $A$ is first order and $A^* M^{-1} A$ is simply the $L^2$ inner product $(\operatorname{rot} E, \operatorname{rot} F)$, here $M^{-1}$ genuinely has to be applied. We therefore carry the Riesz representative $w = M^{-1}(L - zM)u$ as a second unknown on $Z = W_h \times V_h$ and let SLEPc eliminate it:

$$\begin{bmatrix} -M_W & R \\ R^* & 0 \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} = \lambda \begin{bmatrix} 0 & 0 \\ 0 & M_V \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix}, \qquad \lambda = \sigma_{\inf}^2, \tag{4}$$

whose first row is $w = M_W^{-1} R u$ and whose second is $R^* M_W^{-1} R u = \lambda M_V u$. In Firedrake this is one `LinearEigenproblem` built from the two forms

$$a\big((w,u),(\psi,v)\big) = -(w,\psi) + \big[-(\Delta u,\psi) + (\partial_\nu u,\psi)_{\partial\Omega} - z(u,\psi)\big] + \big[-(w,\Delta v) + (w,\partial_\nu v)_{\partial\Omega} - \bar{z}\,(w,v)\big],$$
$$b\big((w,u),(\psi,v)\big) = (u,v),$$

in which no mesh size, penalty or Nitsche term appears anywhere. Note that $w$ is the Riesz representative $M^{-1}(L - zM)u$, *not* a first-order splitting of the Laplacian: there is no $\sigma = \nabla u$ here, unlike `folded_poisson.py`, where $A(u, \sigma) = (-i\sigma', -iu')$ is genuinely first order and is what lets that code drop to $C^0$ elements. Here the second block exists only to apply $M^{-1}$; the Laplacian itself is folded whole.

Two practical points. $B$ in (4) is positive *semi*-definite (it is zero on the $w$ block), so the pencil is not a GHEP; it is solved as a GNHEP, and the $\dim W_h$ infinite eigenvalues are mapped to 0 by the shift–invert and never selected. The target must be a finite *negative* number, not 0: the $(2,2)$ block of $A - \tau B$ is $-\tau M_V$, so a tiny $|\tau|$ makes the LU hit zero pivots, whereas $\tau = -1$ leaves the Schur complement $M_V + R^* M_W^{-1} R$, which is SPD for every $z$, including $z$ in the spectrum. Since all $\lambda \ge 0$, the eigenvalue closest to $\tau = -1$ is $\lambda_{\min}$, which is the one we want. One further detail: on strongly graded meshes the Argyris $1/h$, $1/h^2$ dof rescaling makes the entries span several decades and PETSc's LU with the natural ordering hits an exact zero pivot, so we ask for `-st_pc_factor_mat_ordering_type rcm`.

## 2 Exact Neumann spectra

**Unit square.** $\lambda = \pi^2(m^2 + n^2)$, $m, n \ge 0$, eigenfunctions $\cos(m\pi x)\cos(n\pi y)$, all real-analytic. $\Omega$ is convex, so $\mathrm{D}(A_N) = H^2(\Omega) \cap \{\partial_\nu u = 0\}$ and $V_h \subset \mathrm{D}(A_N)$ is conforming.

**Pizza slice.** $\Omega$ is the sector of radius 1 and aperture $\omega = \frac{3}{2}\pi$, as in Appendix A of the spectral theory notes but with Neumann instead of Dirichlet data. Separating $\Phi = R(\rho)\Theta(\theta)$, the angular problem becomes $\Theta'' = -\mu\Theta$ with $\Theta'(0) = \Theta'(\omega) = 0$, hence

$$\Theta_n(\theta) = \cos\Big(\tfrac{2}{3} n\theta\Big), \qquad \mu_n = \Big(\tfrac{n\pi}{\omega}\Big)^2 = \Big(\tfrac{2n}{3}\Big)^2, \qquad n = 0, 1, 2, \ldots, \tag{5}$$

so the Bessel order $\alpha_n = n\pi/\omega = 2n/3$ is the *same* as in the Dirichlet case (there $\Theta_n = \sin(\frac{2}{3}n\theta)$, $n \geq 1$); Neumann only adds $n = 0$. The radial factor is again $R = J_{\alpha_n}(\sqrt{\lambda}\rho)$, and $\partial_\rho R(1) = 0$ now gives $J'_{\alpha_n}(\sqrt{\lambda}) = 0$, so

$$\Phi_{m,n}(\rho,\theta) = J_{\alpha_n}(j'_{\alpha_n,m}\,\rho)\cos\Big(\tfrac{2}{3}n\theta\Big), \qquad \lambda_{m,n} = \big(j'_{\alpha_n,m}\big)^2, \tag{6}$$

$j'_{\alpha,m}$ being the $m$-th positive zero of $J'_\alpha$ (together with $\lambda = 0$, the constant). Contrast the Dirichlet answer $\lambda_{m,n} = (j_{\alpha_n,m})^2$ of Appendix A.

| $\lambda$ | $n$ | $m$ | $\alpha_n$ | regularity of $\Phi_{m,n}$ |
|---|---|---|---|---|
| 0.0000000 | 0 | 0 | 0 | analytic |
| 1.9634119 | 1 | 1 | 2/3 | $H^{5/3-\epsilon}$, **not** $H^2$ |
| 5.0974588 | 2 | 1 | 4/3 | $H^{7/3-\epsilon}$ |
| 9.3283632 | 3 | 1 | 2 | analytic |
| 14.6170406 | 4 | 1 | 8/3 | $H^{11/3-\epsilon}$ |
| 14.6819706 | 0 | 1 | 0 | analytic |
| 20.9387920 | 5 | 1 | 10/3 | $H^{13/3-\epsilon}$ |
| 23.5286590 | 1 | 2 | 2/3 | $H^{5/3-\epsilon}$, **not** $H^2$ |

Since $J_\alpha(k\rho) \sim \rho^\alpha$ at the origin, $\Phi_{m,n} \in H^{1+2n/3-\epsilon}(\Omega)$, analytic exactly when $2n/3 \in \mathbb{Z}$. The eigenfunctions that fail to lie in $H^2$ are precisely those with $0 < \alpha_n < 1$, i.e. $n = 1$: the entire family

$$\lambda_{m,1} = (j'_{2/3,m})^2 = 1.9634119,\ 23.5286590,\ 64.6642589, \ldots \tag{7}$$

with $\Phi_{m,1} \in H^{5/3-\epsilon}$ and no better. (Note $5/3 = 1 + \pi/\omega$, not $5/2$; an exponent $5/2$ would need $\pi/\omega = 3/2$, i.e. a convex corner $\omega = 2\pi/3$.) The lowest of them is already the *first* nonzero eigenvalue.

## 3 Why the pizza slice must fail

Write $\mathrm{D}(A_N)$ with the graph norm $\|u\|_A^2 = \|u\|_{L^2}^2 + \|\Delta u\|_{L^2}^2$ and put

$$W := H^2(\Omega) \cap \{\partial_\nu u = 0\} \subset \mathrm{D}(A_N). \tag{8}$$

By Grisvard, on a domain with one re-entrant corner of aperture $\omega \in (\pi, 2\pi)$ one has the topological direct sum $\mathrm{D}(A_N) = W \oplus \mathrm{span}\{S\}$ with $S \simeq \rho^{\pi/\omega}\cos(\pi\theta/\omega)$; in particular $W$ is a *closed* subspace of $\mathrm{D}(A_N)$ of codimension one.

**Proposition 1.** *Let $\lambda$ be a simple eigenvalue of $A_N$ whose eigenfunction $\varphi$ is not in $H^2$. Then $\gamma_W(\lambda) := \inf\{\|(A_N - \lambda)u\|_{L^2} : u \in W,\ \|u\|_{L^2} = 1\} > 0$.*

*Proof.* Suppose $u_k \in W$, $\|u_k\| = 1$, $\|(A_N - \lambda)u_k\| \to 0$. Split $u_k = c_k\varphi + r_k$ with $r_k \perp \varphi$. By the spectral theorem $\|(A_N - \lambda)u_k\| = \|(A_N - \lambda)r_k\| \geq \delta\,\|r_k\|$ with $\delta = \mathrm{dist}(\lambda, \sigma(A_N) \setminus \{\lambda\}) > 0$, so $r_k \to 0$ in $L^2$ and $u_k \to \pm\varphi$ in $L^2$. Then $\Delta u_k = -\lambda u_k - (A_N - \lambda)u_k \to -\lambda\varphi$ in $L^2$ too, so $u_k \to \pm\varphi$ in the graph norm. As $W$ is graph-closed, $\varphi \in W \subset H^2$ — a contradiction. □

Any $H^2$-conforming space satisfies $V_h \subset H^2$, and the dual norm (3) forces $\partial_\nu u \to 0$; the discrete problem therefore $\Gamma$-converges to the minimisation over $W$, not over $\mathrm{D}(A_N)$, and Proposition 1 says the singular eigenvalues cannot be detected. Note that this is a statement about the *space*, not about resolution: refining the mesh, or grading it toward the corner, does not help at all, because $W$ is closed.

What the method converges to instead is

$$\sigma^h_{\inf}(z) \ \longrightarrow\ \mathrm{dist}\big(z,\ \sigma(A_N) \setminus \{\lambda_{m,1}\}\big), \tag{9}$$

i.e. the sawtooth of the spectrum *with the singular eigenvalues deleted.*

## 4 What the computations show

**Unit square.** $\sigma^h_{\text{inf}}(z)$ reproduces $\operatorname{dist}(z, \sigma(A_N))$ over $z \in [0, 45]$ with maximum error $2.75 \cdot 10^{-4}$ ($N = 6$), $5.66 \cdot 10^{-6}$ ($N = 10$), $1.54 \cdot 10^{-7}$ ($N = 16$) — about $O(h^{7.6})$, consistent with $\sigma^2_{\text{inf}}$ being an eigenvalue of a quintic $H^2$-conforming discretisation. No boundary condition is imposed anywhere in the code. See `output/of_neumann_argyris/square.pdf`.

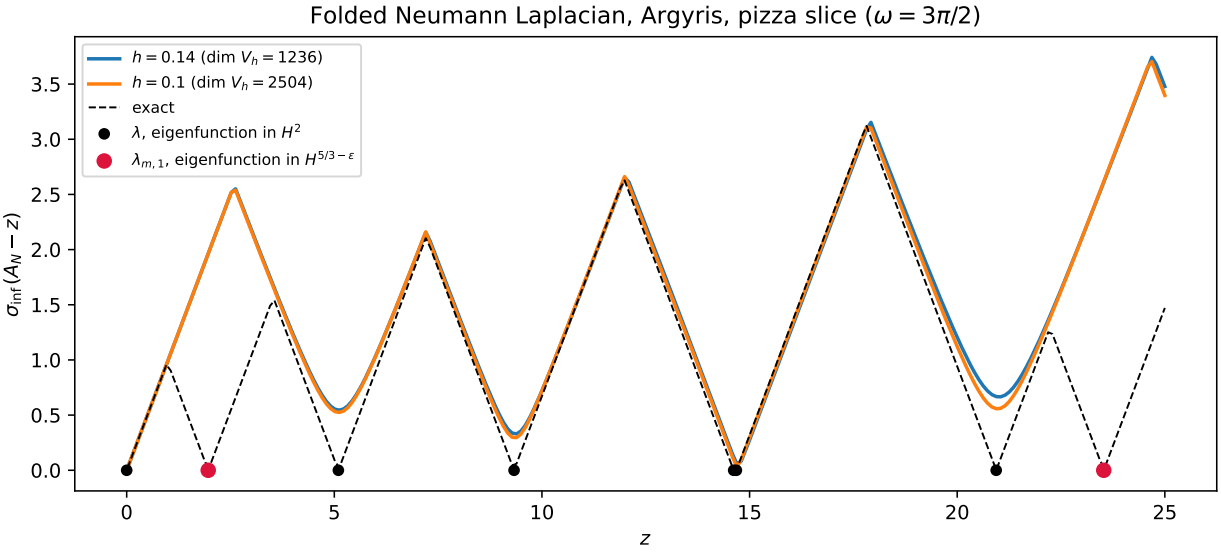
**Pizza slice.** The smooth modes behave, the singular ones do not. Below, $\sigma^h_{\text{inf}}$ at exact eigenvalues, where the true value is 0 in every column; the last two rows grade the mesh toward the re-entrant corner.

| | | singular, $\alpha = 2/3$ | | in $H^2$ | | |
|---|---|---|---|---|---|---|
| $h$ | $h_{\min}$ | 1.963412 | 23.528659 | 5.097459 ($\alpha = 4/3$) | 9.328363 (analytic) | 14.681971 (analytic) |
| 0.20 | $1.9 \cdot 10^{-1}$ | 1.963412 | 2.570532 | 0.664962 | 0.371178 | 0.117124 |
| 0.14 | $1.3 \cdot 10^{-1}$ | 1.963412 | 2.609765 | 0.545006 | 0.329740 | 0.064499 |
| 0.10 | $9.1 \cdot 10^{-2}$ | 1.963412 | 2.615103 | 0.523946 | 0.292930 | 0.036163 |
| 0.07 | $6.2 \cdot 10^{-2}$ | 1.963412 | 2.617157 | 0.424935 | 0.254975 | 0.020143 |
| 0.15 | $2.7 \cdot 10^{-3}$ | 1.963412 | 2.598933 | 0.248126 | 0.343742 | 0.075959 |
| 0.15 | $1.9 \cdot 10^{-4}$ | 1.963412 | 2.607819 | 0.219641 | 0.334596 | 0.074654 |
| limit (9) | | **1.963412** | **2.589867** | 0 | 0 | 0 |

Read this table three ways.

- $\alpha = 2/3$ **(not in $H^2$).** $\sigma^h_{\text{inf}}$ sits on the predicted floor (9) and does not move: at $z = 1.963412$ it agrees with $\operatorname{dist}(z, \sigma(A_N) \setminus \{\lambda_{m,1}\}) = 1.9634119$ to every digit printed, on all six meshes; at $z = 23.528659$ it agrees with 2.589867 to about 0.4%. Grading the mesh toward the corner by four orders of magnitude in $h_{\min}$ changes nothing. *These eigenvalues are simply not in the spectrum that the method sees.*
- $\alpha = 4/3$ **(in $H^2$, but only $H^{7/3-\epsilon}$).** Captured, but slowly, and corner grading helps a lot ($0.425 \to 0.220$): this is an ordinary resolution problem, exactly the behaviour one expects for a function that *is* in the space being approximated badly.
- **analytic modes.** Captured. The purely radial one ($\alpha = 0$) converges cleanly, $0.117 \to 0.020$, roughly $O(h)$.

The contrast between the first two bullets is the whole point: grading fixes the $H^2$ mode and does nothing at all for the non-$H^2$ one, because the obstruction is the closed subspace of Proposition 1 and not the mesh.



In the figure the black markers are the eigenvalues whose eigenfunctions lie in $H^2$ — the computed curves touch zero at every one of them — and the two red markers are $\lambda_{1,1}$ and $\lambda_{2,1}$, where the computed curves ignore the notch in the exact (black dashed) distance function entirely.

**A caveat on the geometry.** The arc is meshed by straight-sided triangles, so $\partial_\nu \varphi = O(h)$ on the chords and the boundary term in (3) limits the *analytic* modes to $O(\sqrt{h})$; the purely radial modes ($n \equiv 0$) escape this and converge like $O(h)$. Curving the geometry would remove it, but Firedrake's Argyris transformation evaluates the Jacobian at the cell barycentre and so is only valid on affine cells. This is a nuisance, not the phenomenon: it affects the modes that *are* captured, not the ones that are missed.